Resilience Academy

SDI System Installation & Customization with CRD and KoboToolbox

Theme 2: Climate Risk Database (CRD)

Architecture, Installation, Configuration & Customization

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Day 2 of 4 | crd.resilienceacademy.ac.tz

Resilience Academy

Theme 2 Overview

Goal: Install, configure and customise a full CRD instance

By end of Day 2 you have a running local CRD with Tanga and Mwanza datasets loaded.

Morning

- CRD/GeoNode architecture deep-dive
- Live walkthrough: crd.resilienceacademy.ac.tz
- Practical 2.1: CRD Docker installation (3h)
- Practical 2.2: CRD Ubuntu bare-metal (3h)

Afternoon

- Practical 2.3: CRD customisation (2h)
- Practical 2.4: Load Tanga & Mwanza data (1h)
- QGIS: WMS/WFS connections to live CRD
- Q&A and troubleshooting

Reference Platform

crd.resilienceacademy.ac.tz — Search: `mwanza_businesses`, `mwanza_education_facilities`, `Tanga_imagery_reduced`

GeoNode Technology Stack

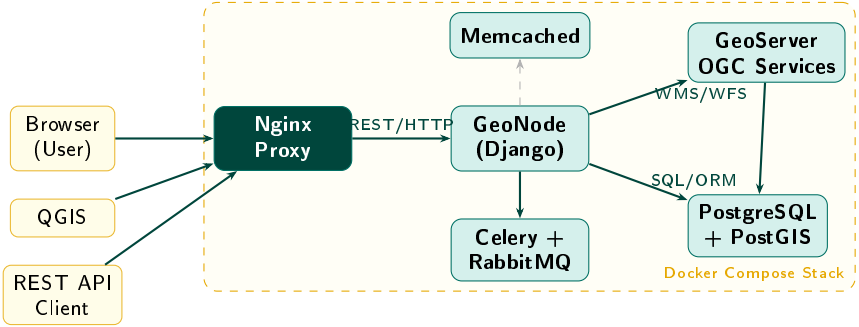
What is GeoNode?

- Open-source geospatial CMS
- CRD = GeoNode customised for Tanzania climate risk data
- Powers 287+ datasets at crd.resilienceacademy.ac.tz
- OGC-compliant: every layer = WMS + WFS + WCS endpoint

Technology Stack

Layer	Technology
Frontend	Django + React
Map server	GeoServer
Database	PostgreSQL + PostGIS
Cache	Memcached
Task queue	Celery + RabbitMQ
Proxy	Nginx
Containers	Docker Compose

CRD Architecture — Data Flow



OGC Services: automatically available for every uploaded layer

WMS (rendered maps) WFS (vector download) WCS (raster) WMTS (tiles) CSW (metadata)

Tanga & Mwanza Datasets on Live CRD

Mwanza Datasets

- `geonode:mwanza_businesses` — exposure points
- `geonode:mwanza_offices` — exposure points
- `geonode:mwanza_education_facilities` — schools
- Mirongo River flood data — hazard layer

Tanga Datasets

- `geonode:Tanga_imagery_reduced` — 50cm KOMPSAT-3 satellite imagery (2014)
- Tanga flood mapping layers (GIZ/RA)
- Export Processing Zone wastewater data

Search on Live CRD

https://crd.resilienceacademy.ac.tz/catalogue/#/search?q=mwanza
https://crd.resilienceacademy.ac.tz/catalogue/#/search?q=tanga

CRD Docker Installation

Steps 1-3: Clone, Configure

```
git clone https://github.com/GeoNode/
  geonode-project.git my-crd
cd my-crd
python3 -m venv .venv
source .venv/bin/activate
pip install "Django>=3.2,<4.0"
django-admin startproject \
  --template=. \
  --extension py,rst,json,yml,ini,env \
  my_geonode .
cp .env.sample .env
nano .env
# SITEURL=http://localhost/
# POSTGRES_PASSWORD=resilience2024
# SECRET_KEY=<random_string>
```

Steps 4-6: Build, Run, Init

```
# Build Docker images
docker compose build

# Start all services
docker compose up -d

# Watch startup (3-5 min)
docker compose logs -f

# Initialize
docker compose exec django \
  python manage.py migrate
docker compose exec django \
  python manage.py createsuperuser
docker compose exec django \
  python manage.py sync_geonode_maps

# Access: http://localhost
```

CRD Customisation

local_settings.py

```
SITE_NAME = "Tanzania SDI Training CRD"
SITE_HOST_NAME = "localhost"
THEME_ACCOUNT_CONTACT_EMAIL = \
  "admin@resilienceacademy.ac.tz"

# Default map: Mwanza
DEFAULT_MAP_CENTER = (32.9, -2.5)
DEFAULT_MAP_ZOOM = 12

# Or Tanga:
# DEFAULT_MAP_CENTER = (39.1, -5.07)

DEFAULT_ANONYMOUS_VIEW_PERMISSION = True
```

Download Mwanza Data (WFS)

```
curl -s "https://crd.resilienceacademy.ac.tz
  /geoserver/ows?service=WFS
  &version=2.0.0&request=GetFeature
  &typeName=geonode:mwanza_businesses
  &outputFormat=application/json" \
  -o mwanza_businesses.geojson

# Count features:
python3 -c "
import json
with open('mwanza_businesses.geojson') as f:
  d=json.load(f)
print(len(d['features']),'features')"
```

Day 2 Summary

Achieved

- ✓ GeoNode/CRD architecture understood
- ✓ CRD running locally via Docker
- ✓ CRD installed on Ubuntu bare metal
- ✓ Branding and map settings customised
- ✓ Mwanza and Tanga data loaded
- ✓ Layers validated in QGIS via WMS/WFS

Day 3: KoboToolbox

- KoboToolbox architecture and hosting
- Self-hosted install with kobo-install
- Tanga flood survey form design
- Backend management and backups
- KoboToolbox to CRD integration

Keep your local CRD running!

Q&A and Troubleshooting

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